

Reading the Surface:

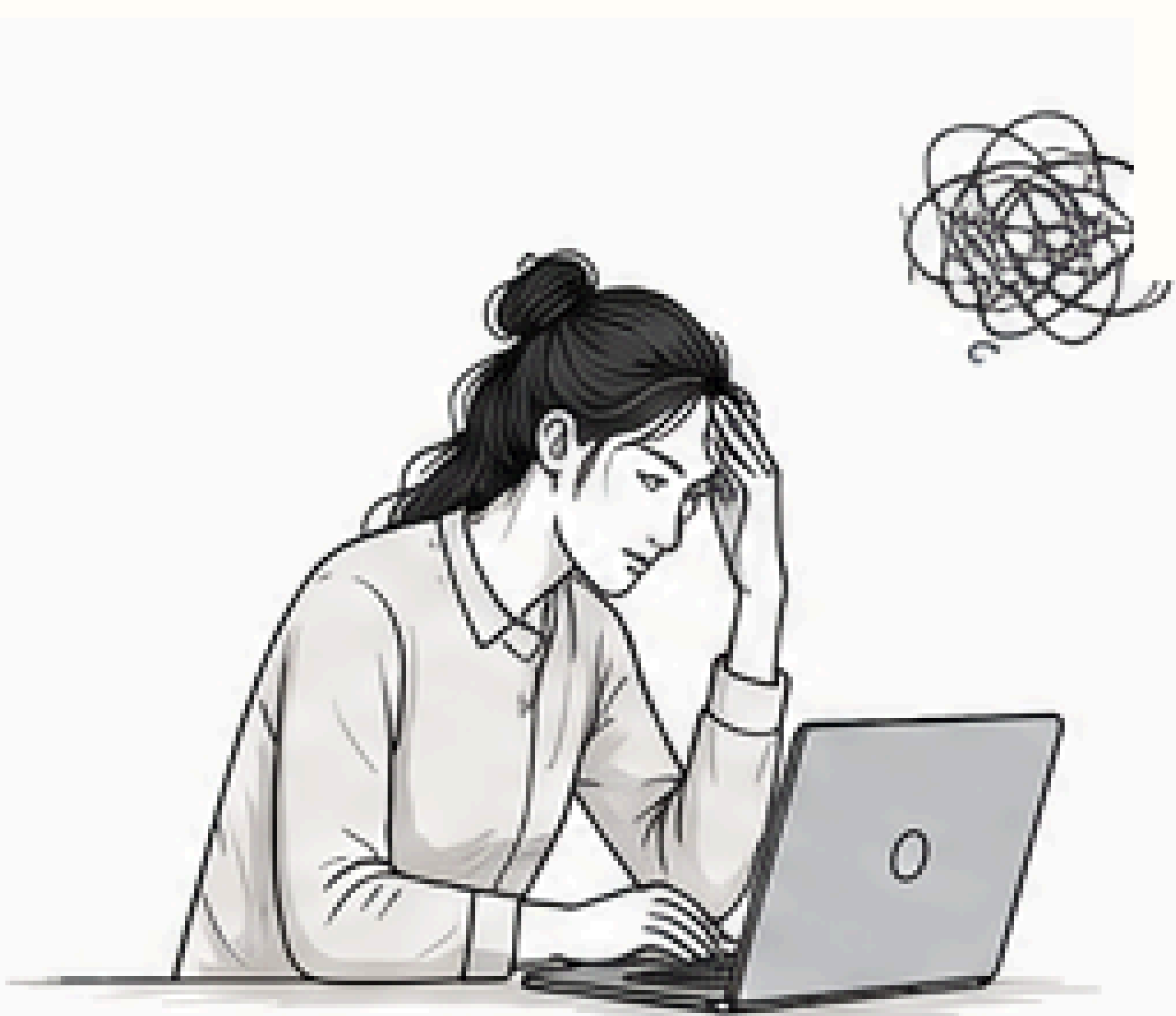
Social Perception under Expressive Compression in AI-Assisted Communication

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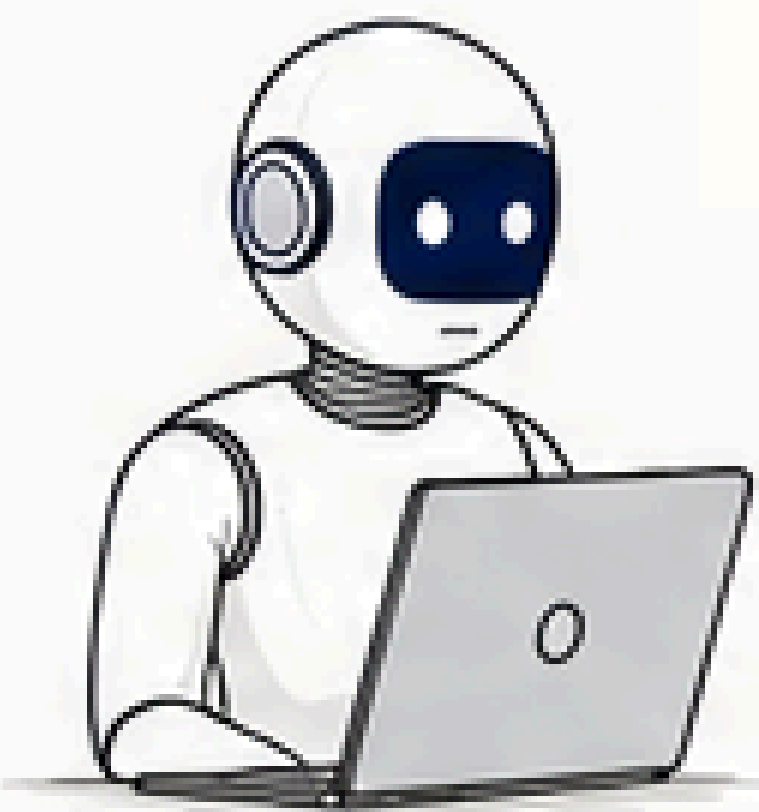
Distressed Writer

Emotions: panic, anger, distrust, frustration



AI-Assisted Rewriting

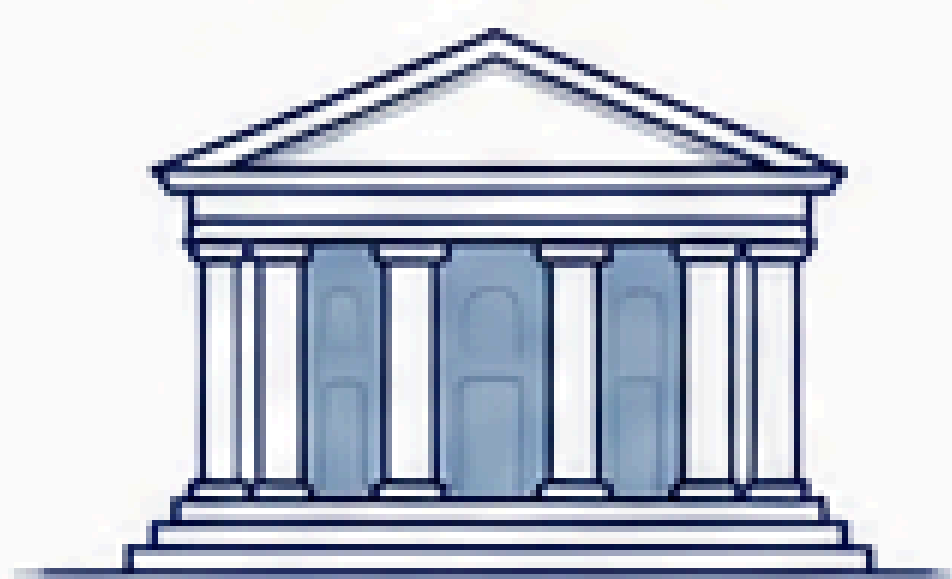
Alignment & safety filters



- ✓ Safety constraints
- ✓ Helpfulness objectives
- ✓ Neutral tone
- ✓ Broadly acceptable style

Institutional Audience

Gatekeeping power, evaluation, risk



Emotional urgency is compressed into calm, institutional language.

1 Problem



Perception depends on context, intent, and constraint—not just surface content.



High-stakes communication is filtered by human strategy and technical systems.

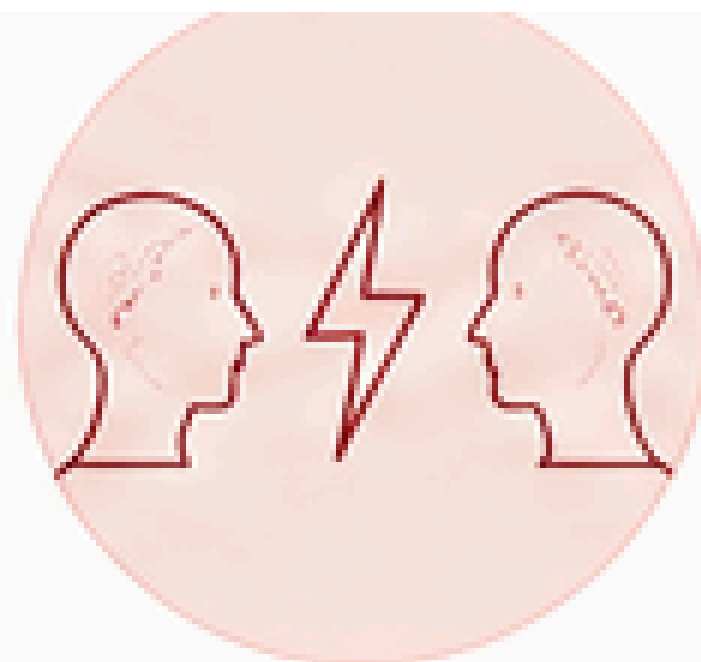


Models risk learning compressed personas rather than underlying situations.

3 Simulated Congruence in AI-Assisted Communication

Underlying Reality

Conflict, power imbalance, mistrust, urgency



Compression

Human compression + Model compression



Surface Output

Polite, concise, reasonable, institutionally appropriate



Simulated Congruence

Texts appear harmonious even when the underlying relationship is tense



Example: Evolution of a Sensitive Message

Raw Draft

This decision is unfair and disastrous. I'm panicked and furious. You're not listening.

Self-Edited

I am concerned about the decision and its impact. I would appreciate a reconsideration.

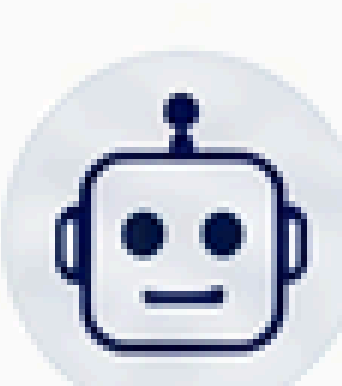
LLM-Polished

I respectfully request reconsideration of the decision in light of the circumstances.

2 Expressive Compression



Writers compress strong emotions into polite, concise, and apparently reasonable language.



Aligned LLMs smooth expression to satisfy safety and helpfulness objectives.



Compression is protective and strategic—but obscures evidence of constraint.

4 Where the Blind Spot Appears



Production of Texts

- Safety and strategy drive self-censorship.
- Tools standardize tone.

Annotation as Data

- Annotators see only polished outputs.
- Labels reflect surfaces, not constraints.

Model Training & Evaluation

- Models learn correlations on compressed data.
- Risk of misjudging urgency, risk, and credibility.

5 Implications for NLP



- Treat production conditions as part of social context.
- Detect or account for expressive compression.
- Incorporate context about power, risk, and stakes.
- Design evaluations that value transparency over style.
- Support tools that make constraint visible when relevant.



Surface calm may be a product of constraint, not evidence of low conflict.

Selected References

Blodgett et al. (2020) • Bender et al. (2021) • Ouyang et al. (2022) • Bai et al. (2022)